

SEMICONDUCTOR ELECTRONICS : MATERIALS , DEVICES AND SIMPLE CIRCUITS

Important formulas and concepts directly from the NCERT-

- The energy bands are the different energy levels with continuous energy variation.
- The energy bands which includes the energy levels of valence electrons k/as '**valence band**'.
- The energy band above the valence band is called the '**conduction band**'.

❖ Classification b/w conductors(metal) , insulators and semiconductors-

1) Metals-

- Energy gap E_g (gap b/w valence band and conduction band) ≈ 0
- As temperature increases, resistivity increases and conductivity decreases.

2) Insulators-

- $E_g > 3\text{eV}$ (large)
- As temp. increases, conductivity increases in very small amount.

3) Semiconductors-

- $E_g < 3\text{eV}$
- **At 0 kelvin, act as insulator.**
- As temp. increases, conductivity increases and act as conductors.

❖ Intrinsic semiconductors-

- In intrinsic semiconductors, the no. of free electrons, n_e is equal to no. of holes, n_h .

$$n_e = n_h = n_i$$

where, n_i is **intrinsic carrier concentration**.

- Hole is vacancy with effective positive charge.
- Total current is the sum of electron current and hole current.

$$I = I_e + I_h$$

- Apart from process of generation of conduction of electrons and holes , a process of recombination occurs in which electrons recombine with the holes.

- At equilibrium, the rate of generation is equal to the rate of recombination of charge carriers.

❖ Extrinsic semiconductors-

- When a small amount of suitable impurity is added to intrinsic semiconductor, its conductivity increases and this material is k/as 'extrinsic semiconductors'.
- The addition of impurity is 'doping' and the impurity atom is called 'dopant'.
- There are 2 types of dopants used in doping forms 2 types of extrinsic semiconductors-

1) **n-type semiconductors-**

- Impurity added is pentavalent (**As,Sb,P etc.**).
- The pentavalent dopant is donating one extra electron for conduction and hence k/as 'donor impurity'.
- In a doped semiconductor, total no. of electrons are due to contributed by donors and those generated intrinsically, while holes due to intrinsic source only.
- Electrons are majority charge carriers and holes are minority charge carriers.
- For n-type semiconductors,

$$n_e \gg n_h$$

2) **p-type semiconductors-**

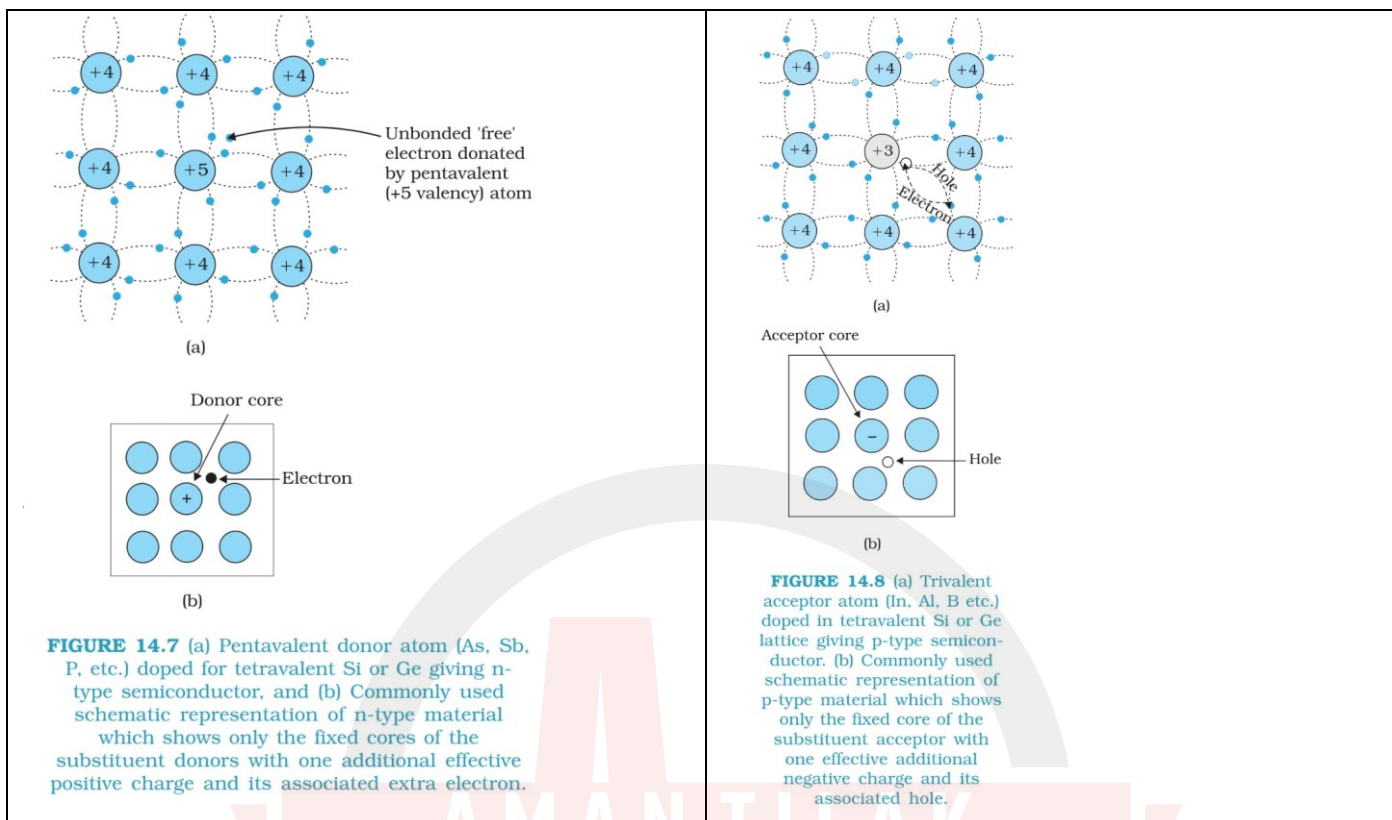
- impurity added is trivalent (**B,Al,In etc.**).
- The trivalent impurity is k/as 'acceptor impurity' because it can generate hole by accepting electron.
- Total no. of holes are more than no. of electrons because holes generated due to dopants as well as by intrinsic source and e^- by intrinsic source only.
- Holes are majority charge carriers and electrons are minority .
- For p-type semiconductors,

$$n_h \gg n_e$$

- The electron and hole concentration in a semiconductor in thermal equilibrium is given by,

$$n_e n_h = n_i^2$$

- The difference in resistivity of C,Si and Ge depends on energy gap.



❖ p-n junction –

- free electrons diffuse from n-side to p-side due to concentration difference and recombine with the hole.
- **Diffusion current flows from P to N side due to majority charge carriers.**
- Due to electron and hole recombination, depletion layer form.
- After formation of depletion layer, there is an internal electric field which allow to move minority charge carriers, due to which **drift current flows from N to P side due to minority charge carriers.**
- At equilibrium,

$$I_{\text{diffusion}} = I_{\text{drift}}$$

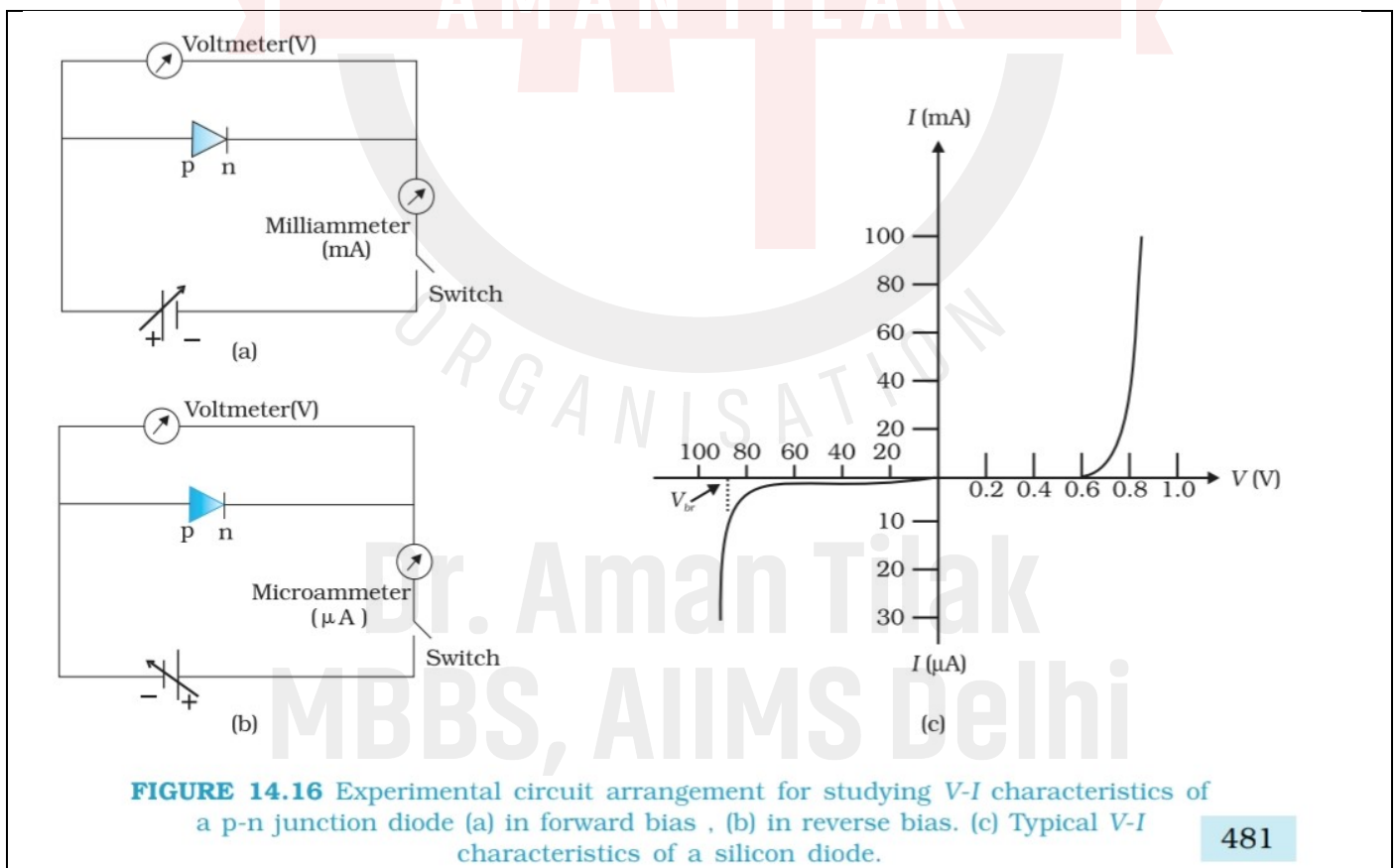
❖ p-n junction diode under forward bias-

- negative terminal of battery is connected to n-side and positive terminal to p-side.
- Net electric field decreases.
- Potential barrier decrease.
- Width of depletion decrease.

- Diffusion current will flow.
- Drift current ≈ 0
- In forward bias, **ideal diode act as simple wire.**

❖ p-n junction diode under reverse bias-

- negative terminal connected to p-side and positive terminal to n-side.
- Net electric field increase.
- Potential barrier increase.
- Width of depletion increase.
- Diffusion current ≈ 0 .
- **Drift current is very small because due to minority charge carriers, k/as ‘reverse saturation current’.**
- In reverse bias, **ideal diode behave as open wire.**
- If voltage is equal to breakdown voltage, current increases sharply.



- For diodes, a quantity called *dynamic resistance* as ratio of small change in voltage to a small change in current.

$$r_d = \Delta V / \Delta I$$

❖ Application of junction diode as a Rectifier-

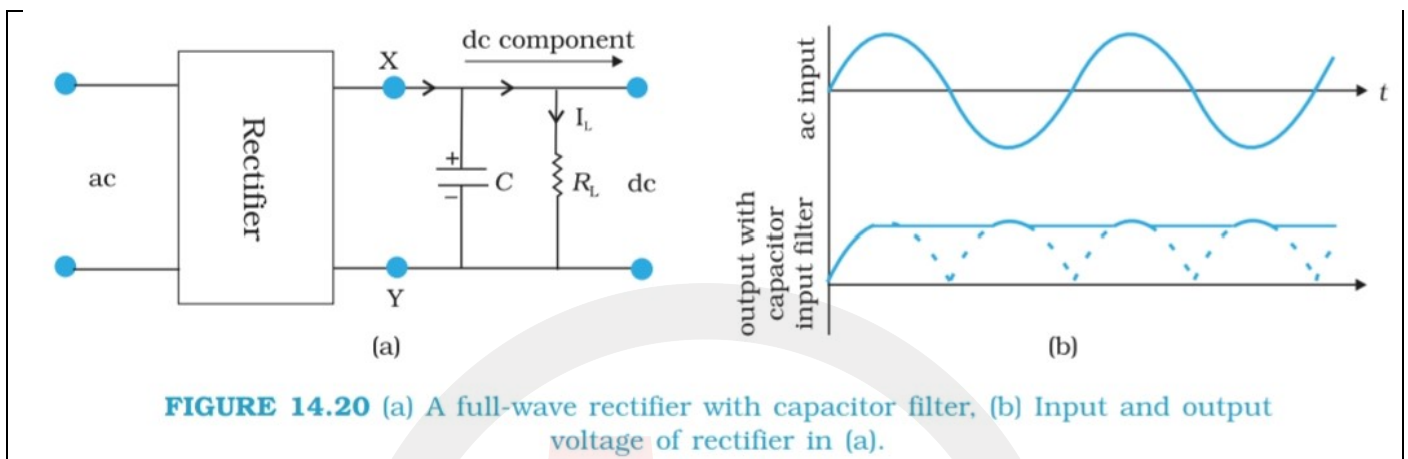
- If an alternating voltage is applied across a diode the current flows only in that part of the cycle when the diode is forward biased. This property is used to rectify alternating voltages and the circuit used for this purpose is called a *rectifier*.
- If an alternating voltage is applied across a diode in series, current will appear only during half cycles, such circuit is known as '**half wave rectifier**'.
- For half wave rectifier, output frequency = input frequency



- The circuit using two diodes as shown will give rectified output to both positive and negative half cycles, hence it is k/as '**full wave rectifier**'.
- In full wave rectifier, output frequency is 2 times of input frequency.



- The additional circuits appear to filter out the ac ripple and give a pure dc voltage, so they are called filters. Capacitor is used as filter in rectifier.



AMANTILAK

ORGANISATION

Dr. Aman Tilak
MBBS, AIIMS Delhi

SUMMARY

1. Semiconductors are the basic materials used in the present solid state electronic devices like diode, transistor, ICs, etc.
2. Lattice structure and the atomic structure of constituent elements decide whether a particular material will be insulator, metal or semiconductor.
3. Metals have low resistivity (10^{-2} to $10^{-8} \Omega\text{m}$), insulators have very high resistivity ($>10^8 \Omega\text{m}^{-1}$), while semiconductors have intermediate values of resistivity.
4. Semiconductors are elemental (Si, Ge) as well as compound (GaAs, CdS, etc.).
5. Pure semiconductors are called 'intrinsic semiconductors'. The presence of charge carriers (electrons and holes) is an 'intrinsic' property of the material and these are obtained as a result of thermal excitation. The number of electrons (n_e) is equal to the number of holes (n_h) in intrinsic conductors. Holes are essentially electron vacancies with an effective positive charge.
6. The number of charge carriers can be changed by 'doping' of a suitable impurity in pure semiconductors. Such semiconductors are known as extrinsic semiconductors. These are of two types (n-type and p-type).
7. In n-type semiconductors, $n_e \gg n_h$ while in p-type semiconductors $n_h \gg n_e$.
8. n-type semiconducting Si or Ge is obtained by doping with pentavalent atoms (donors) like As, Sb, P, etc., while p-type Si or Ge can be obtained by doping with trivalent atom (acceptors) like B, Al, In etc.
9. $n_e n_h = n_i^2$ in all cases. Further, the material possesses an *overall charge neutrality*.
10. There are two distinct band of energies (called valence band and conduction band) in which the electrons in a material lie. Valence band energies are low as compared to conduction band energies. All energy levels in the valence band are filled while energy levels in the conduction band may be fully empty or partially filled. The electrons in the conduction band are free to move in a solid and are responsible for the conductivity. The extent of conductivity depends upon the energy gap (E_g) between the top of valence band (E_v) and the bottom of the conduction band E_c . The electrons from valence band can be excited by

heat, light or electrical energy to the conduction band and thus, produce a change in the current flowing in a semiconductor.

11. For insulators $E_g > 3$ eV, for semiconductors E_g is 0.2 eV to 3 eV, while for metals $E_g \approx 0$.
12. p-n junction is the 'key' to all semiconductor devices. When such a junction is made, a 'depletion layer' is formed consisting of immobile ion-cores devoid of their electrons or holes. This is responsible for a junction potential barrier.
13. By changing the external applied voltage, junction barriers can be changed. In forward bias (n-side is connected to negative terminal of the battery and p-side is connected to the positive), the barrier is decreased while the barrier increases in reverse bias. Hence, forward bias current is more (mA) while it is very small (μ A) in a p-n junction diode.
14. Diodes can be used for rectifying an ac voltage (restricting the ac voltage to one direction). With the help of a capacitor or a suitable filter, a dc voltage can be obtained.

POINTS TO PONDER

1. The energy bands (E_C or E_V) in the semiconductors are space delocalised which means that these are not located in any specific place inside the solid. The energies are the overall averages. When you see a picture in which E_C or E_V are drawn as straight lines, then they should be respectively taken simply as the *bottom* of conduction band energy levels and *top* of valence band energy levels.
2. In elemental semiconductors (Si or Ge), the n-type or p-type semiconductors are obtained by introducing 'dopants' as defects. In compound semiconductors, the change in relative stoichiometric ratio can also change the type of semiconductor. For example, in ideal GaAs the ratio of Ga:As is 1:1 but in Ga-rich or As-rich GaAs it could respectively be $\text{Ga}_{1.1}\text{As}_{0.9}$ or $\text{Ga}_{0.9}\text{As}_{1.1}$. In general, the presence of defects control the properties of semiconductors in many ways.